DHANVANTH KUMAR GUDE

Vijayawada, Andhra Pradesh, India | Email | LinkedIn | Scholar | ORCID

Research Interests

- Agentic Artificial Intelligence
- Privacy-Preserving Machine Learning
- Differential Privacy
- Large Language Model Security
- AI Safety and Trustworthy AI

Education

Chandigarh University, India (Aug 2022 – Jun 2026)
B.E. (Hons.) in Computer Science and Engineering
Specialization: Artificial Intelligence and Machine Learning

- **GPA: 8.62/10.00**
- **Research Component**
  - **Capstone Project I (Semester VII):** 90%
  - **Internship (Semester VIII):** 100%
- **Relevant Coursework**: Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing, Probability and Statistics

Publications

Patent

1. **A Method and a System for Privacy-Preserving Authentication and Attestation in Decentralized IoT Swarms**
   Published Indian Patent **Application No.: 202631037175**
   Publication Date: 05 June 2026
   Applicant: Indian Institute of Technology Guwahati
   IPC Classification: G06F 21/57, H04L 9/32, H04L 29/06, H04L 9/08, G06F 21/44
   Inventors: Dr. Satyajit Das, **Dhanvanth Kumar Gude**, Dr. Tamarapalli Venkatesh, Dr. John Jose, and Dr. Ruchika Gupta
   Contribution: Co-Inventor (Contributed to the design and development of the proposed privacy-preserving authentication and attestation system.)

Journal Articles

1. **Transforming Urban Sanitation: Enhancing Sustainability through Machine Learning-Driven Waste Processing**
   Dhanvanth Kumar Gude, Harshavardan Bandari, Anjani Kumar Reddy Challa, Sabiha Tasneem, Zarin Tasneem, Shyama Barna Bhattacharjee, Mohit Lalit, Miguel Angel López Flores, and Nitin Goyal.
   Sustainability (MDPI), 2024
   Contribution: First Author **Q1 | IF: 4.1 | SCIE Indexed**
   DOI

2. **IntelliGuard: An Energy-Efficient and Secure Fog-Integrated UAV Surveillance Framework Enhanced by DPAFIO**
   Dhanvanth Kumar Gude, Vamshi Krishna Raavi, Mohit Lalit Anurag Jain, and Bhupesh Dewangan.
   Discover Computing (Springer Nature), 2026
   Contribution: First Author **Q2 | IF: 1.9 | SCIE Indexed**
   Included in the Springer Nature Topical Collection: *Augmenting Edge AI with Collaborative Edge Computing: From Trusted Nodes to Ubiquitous Personal Devices*.
   DOI

Journal Articles Under Review

1. **GASE: A Graph-Augmented Stacked Ensemble Framework for Suspicious User Threat Detection in Electronic Health Record Systems**
   Adil Hussain Seh, **Dhanvanth Kumar Gude**, Will Hurst.
   Array (Elsevier), Under Review **Q1 | IF: 5.3 | ESCI Indexed**
   Contribution: Co-Author (Algorithm implementation, experimentation, and manuscript preparation)
   Manuscript Number: ARRAY-D-26-02459

2. **A Resilient and Latency-Aware Fog Node Deployment Framework Using Voronoi-Guided Oceanic Swarm Optimization for Large-Scale IoT-Edge Networks**
Dhanvanth Kumar Gude, Adil Hussain Seh, Khaled Alomar, Eesa Alsolami, Mohit Lalit, Kemal Polat, and Anurag Jain.
Journal of Network and Computer Applications (Elseveir), Submitted (July 2026)
Contribution: First Author **Q1 | IF: 7.7 | SCIE Indexed**
Manuscript Number: JNCA-D-26-02990

Conference Publications

1. **Federated Learning Enhanced IoT based Adaptive Lighting System with Pedestrian Detection**.
Dhanvanth Kumar Gude, Vamshi Krishna Raavi, Amandeep Singh Arora, Gyanendra Kumar Shukla, Mohit Lalit, Amit Sandhu
7th International Symposium on Advanced Electrical and Communication Technologies (ISAECT) 2025
Publisher: IEEE
Contribution: First Author
DOI
2. **Privacy-Preserving Dynamic Pricing for EV Charging via Fog-Enabled Federated Learning**.
Dhanvanth Kumar Gude, Vamshi Krishna Raavi, Mohit Lalit, Nitin Goyal
7th International Conference on Computing, Communication & Automation (ICCCA) 2025
Publisher: IEEE
Contribution: First Author
DOI
3. **Design and Simulation of an IoT-Based Adaptive Lighting System with Pedestrian Detection**.
Vamshi Krishna Raavi, **Dhanvanth Kumar Gude**, Davinder Singh, Mohit Lalit, Amit Sandhu, Shailja Gera
2025 Internation Conference on Innovations and Emerging Technologies In AI & Communication Systems (IETACS)
Publisher: IEEE
Contribution: Co-Author (Simulation development, experimental evaluation, and performance analysis.)
DOI
4. **A Secure Aggregation-Based Comparative Analysis of Federated Learning Algorithms on Human Activity Recognition Data**.
Harshavardan Bandari, **Dhanvanth Kumar Gude**, Vamshi Krishna Raavi, Anjani Kumar Reddy Challa, Siddharth Kumar
2025 2nd Global AI Summit - International Conference on Artificial Intelligence and Emerging Technology (AI Summit)
Publisher: IEEE
Contribution: Co-Author (Federated learning implementation, comparative analysis, and experimental validation.)
DOI
5. **Human Activity Recognition Reimagined: The CGRNHAR Self-supervised Approach**.
Anjani Kumar Reddy Challa, Vamshi Krishna Raavi, Harshavardan Bandari, **Dhanvanth Kumar Gude**, Siddharth Kumar
2025 2nd Global AI Summit - International Conference on Artificial Intelligence and Emerging Technology (AI Summit)
Publisher: IEEE
Contribution: Co-Author (Experimental evaluation, performance analysis, and manuscript preparation.)
DOI
6. **An Efficient Privacy-Aware Framework for Traffic Violation Detection Using Federated Learning**.
Dhanvanth Kumar Gude, Vamshi Krishna Raavi, Mohit Lalit, Nitin Goyal
2026 1st International Conference on Advancing Sustainable Solutions through Technologies (ICASST)
Publisher: IEEE
Contribution: First Author
DOI
7. **FA-CFL: A Fog-Aware Capacitated Facility Location Framework for Scalable Fog Node Deployment**.
Dhanvanth Kumar Gude, Vamshi Krishna Raavi, Anjani Kumar Reddy Challa, Harshavardan Bandari, Siddharth Kumar
3rd International Conference on Recent Advances in Artificial Intelligence, Computer Vision & Smart Systems 2026
Publisher: AIP conference proceedings
Contribution: First Author
Status: Accepted and Presented

Book Chapter

1. **Applications of Deep Generative AI in Neuroprosthetics**.
Dhanvanth Kumar Gude, Vamshi Krishna Raavi, Mohit Lalit, Sakshi Mehta, Mohit Angurala
Neuroprosthetics Revolution: Transforming Lives with Deep Generative AI
Publisher: Apple Academic Press
Contribution: First Author
DOI

English Language Proficiency

IELTS Academic: Overall Band **7.5**

Component Scores: Listening: **8.0** Reading: **8.0** Writing: **7.0** Speaking: **6.5**

Research Experience

Assistant Project Engineer, Indian Institute of Technology Guwahati (Jun 2026 – Present)

- Currently conducting research under the supervision of **Dr. Tamarapalli Venkatesh** and **Dr. Satyajit Das**, Department of Computer Science and Engineering, Indian Institute of Technology (IIT) Guwahati.
- Designed a **post-quantum, privacy-preserving authentication and attestation protocol called PRISM** for resource constrained IoT devices, extending the **ZKSWARMv1** framework using lightweight cryptographic techniques.
- Developing an **anomaly detection framework** to identify multi-hop protocol laundering attacks in IoT ecosystems using **Cross-Protocol Temporal Coalition Graphs**.
- Research is supported by the **Ministry of Electronics and Information Technology (MeitY), Government of India**.

Research Intern, Indian Institute of Technology Guwahati (Sep 2025 – May 2026)

- Conducted undergraduate research under the supervision of **Dr. Tamarapalli Venkatesh** and **Dr. Satyajit Das**, focusing on **privacy-preserving authentication and attestation for IoT edge networks using zero-knowledge proofs**.
- Designed and implemented **ZKSWARMv1**, a lightweight zero-knowledge-based authentication and attestation framework integrating firmware, sensor, range, and ring proofs, achieving deployment within a **128 KB** memory footprint on resource-constrained platforms (ESP32).
- Published an **Indian Patent** for **ZKSWARMv1** (Application No. **202631037175**), recognizing its novel approach to secure, lightweight, and privacy-preserving IoT authentication.

Teaching Experience

Teaching Assistant, Indian Institute of Technology Guwahati | Odd Semester 2026 (July 2026 – Present)

- Selected as a Teaching Assistant for CS3105 – Operating Systems and CS3105L – Operating Systems Laboratory at IIT Guwahati for the Odd Semester 2026.
- Assisting the course instructor with tutorials, laboratory sessions, assignment evaluation, and academic support for undergraduate students.
- Facilitating student learning by providing guidance on operating systems concepts, Linux programming, process management, memory management, synchronization, and file systems.

Honors & Awards

- **IEEE Luminary of Discover (Best Research) Awardee**, awarded for research excellence by IEEE Chandigarh University Student Branch in IEEE CUSB Excellence Awards 2.0 (2026)
- **Alstom Scholarship Awardee** (2×), awarded for academic excellence and research potential

Skills

Programming: Python, C++, SQL

Artificial Intelligence: Machine Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing

Privacy & AI Security: Privacy-Preserving Machine Learning, Differential Privacy, AI Security, Information Leakage Analysis, Privacy-Aware AI Systems

Agentic AI & Intelligent Systems: Agentic AI, Multi-Agent Systems, Memory-Augmented AI, AI Reasoning, AI Safety and Trustworthy AI

AI Frameworks & Tools: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, Linux, Git

Research & Analytical Skills: Experimental Design, Literature Review, Statistical Analysis, Technical Writing, Problem Formulation